LEV OLEYNIK
EEC 146
In-Person Lab: Photolitography and Characterization
1/26/2026

Photolitography Process Steps:

1.) Clean Wafers
- Chemically, with DI Water, HF acid, to ensure fully clean wafer

2.) Barrier Layer formation
- usually SiO_2, or Si_3N_4, polysilicon, photoresist. Used as barriers.

3.) Photoresist application
- Is light-sensitive, so you can do UV-etching. Photoresist is typically applied in liquid form, applied via fast spinning in a vacuum.

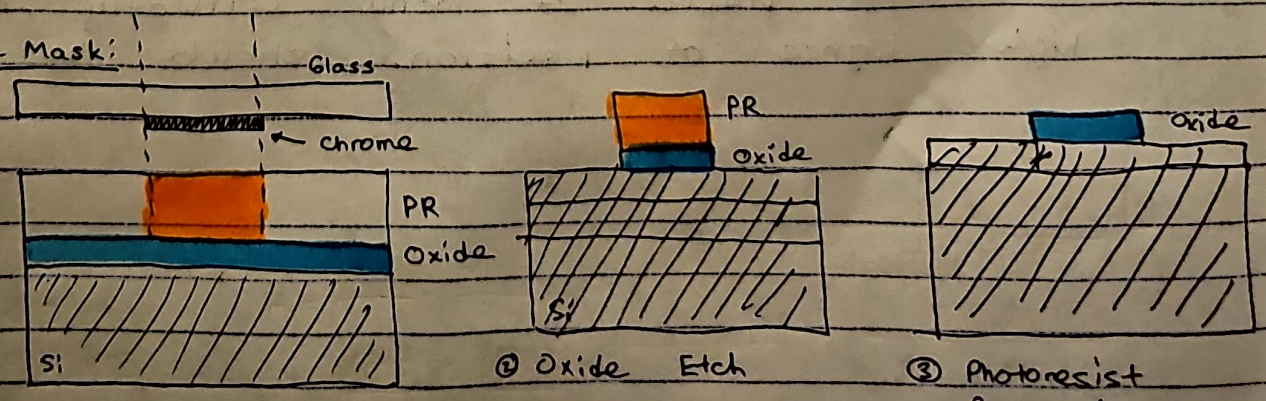
4.) Soft Baking
- Improves adhesion, removes solvent from photoresist (to harden it).

5.) Mask Alignment.
- Done to ensure mask etch is done correctly and you remove what you actually want. Crucial step, is usually done automatically in newer fabs.

6.) Photoresist Exposure and Development
- Can be positive or negative, idea is you can expose and then wash off unwanted photoresist.

7.) Hard Baking
Hardens photoresist, improves adhesion to substrate. Time and temperature can vary.

Positive Mask:

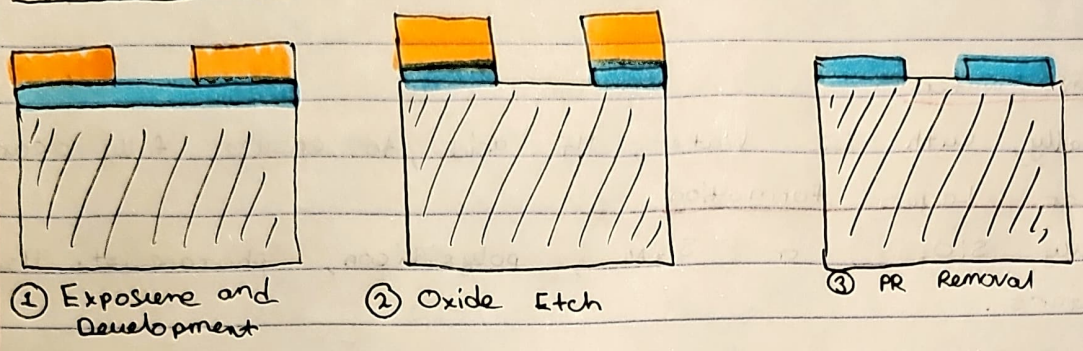


① Exposure and Development

② Oxide Etch

③ Photoresist Removal

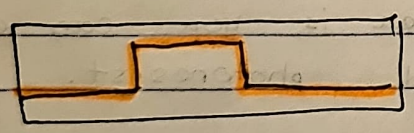
Negative Mask:



▷ Stylus-Based Profilometer:

contact instrument that uses a microscopic stylus to physically trace a surface. Can be used for det. roughness, step heights, & film thickness.
In terms of operation itself, it operates electrochemically, and in the context of our device fabrication it can be used to verify that our steps worked as intended, and our device cut-outs we made are of apropriate widths and depths.

A plot sketch would look like a pulse:



Errors could come from wafer not being perfectly flat, vibration, noise, or Stylus tip radius being not super precise. If done correctly, however, a high degree of precision should be achieved. ✓